

Session 1 homework — MILK10k exploratory data analysis

Computer Vision and Speech Recognition · Stanislaus Lattorff

GitHub repository

Public repository (my personal account):

<https://github.com/stanislauslattorff/milk10k-computer-vision>

- EDA notebook: `notebooks/01_eda_milk10k.ipynb`, saved with all outputs so the plots show on GitHub without running it.
- Code it uses: `src/milk10k/`. Figures it produces: `reports/figures/`.
- Dataset is not in the repo (345 MB, CC BY-NC) — `scripts/download_data.sh` pulls it from ISIC.

The clinical task

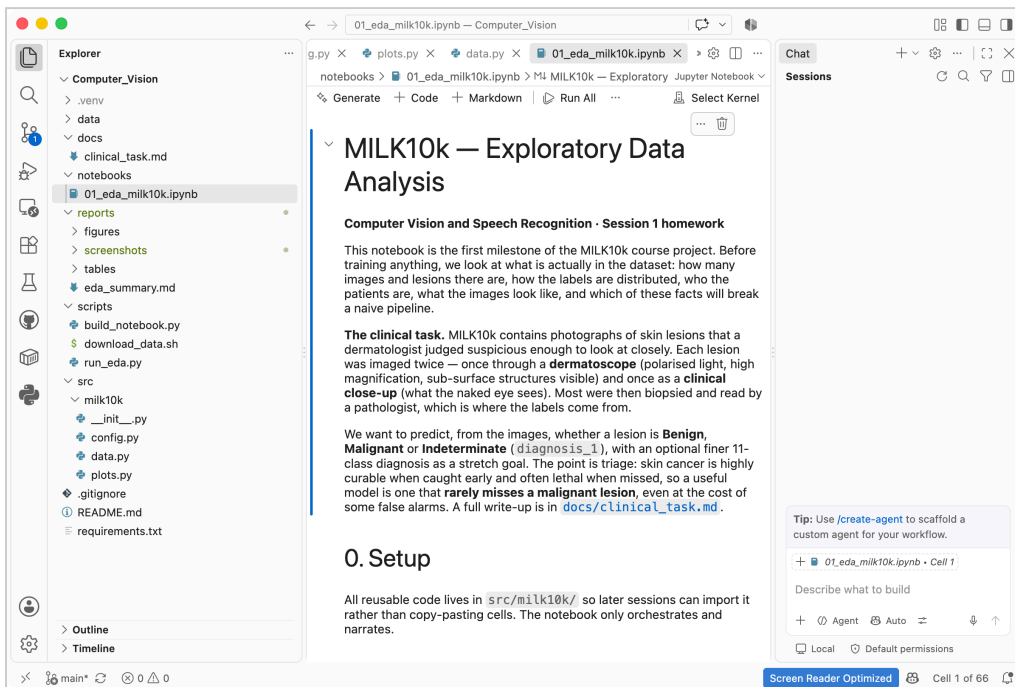
- Photos of skin lesions from patients who were sent to a dermatologist.
- Each lesion is photographed twice: a normal close-up, and one through a dermatoscope (a lens with polarised light that shows structures below the skin surface).
- Almost all lesions were biopsied afterwards, so the labels come from a pathologist and not from someone's opinion.
- Task: predict benign / malignant / indeterminate from the image (`diagnosis_1`). There is a finer 11-class label as an optional target.
- It is a triage tool. Skin cancer found early is removed in a small operation; found late it is a much bigger problem and melanoma can be fatal.
- So missing a cancer is far worse than doing one unnecessary biopsy. The number that matters is recall on the malignant class, not accuracy.
- Important caveat: the dataset only has lesions a doctor already found suspicious enough to biopsy. 69% are malignant, nothing like the general population. A model trained here answers "given a specialist was already worried, is this cancer?" — not "is this cancer?".

What I did

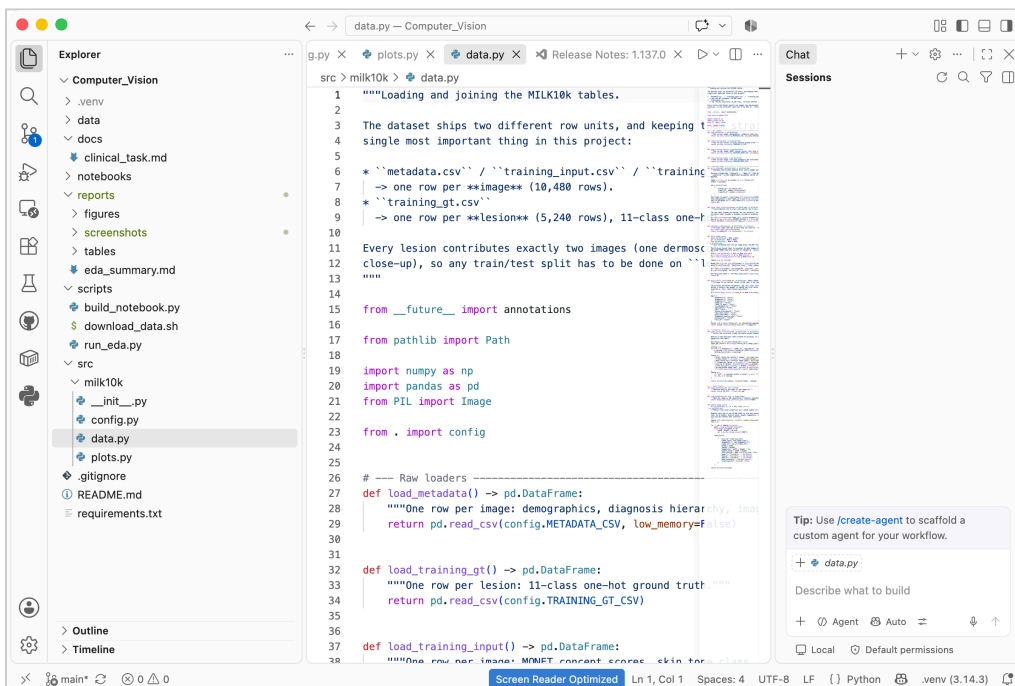
- Downloaded the full dataset: 10,480 images, 5,240 lesions.
- Loaded `metadata.csv` and `training_gt.csv` with pandas.
- Checked the structure instead of assuming it.
- Plotted class distribution, missing values, age, sex, body site, skin tone, image sizes and colour statistics.
- Looked at the actual images, per class and per lesion.

Code editor with the project structure

I am using Visual Studio Code.



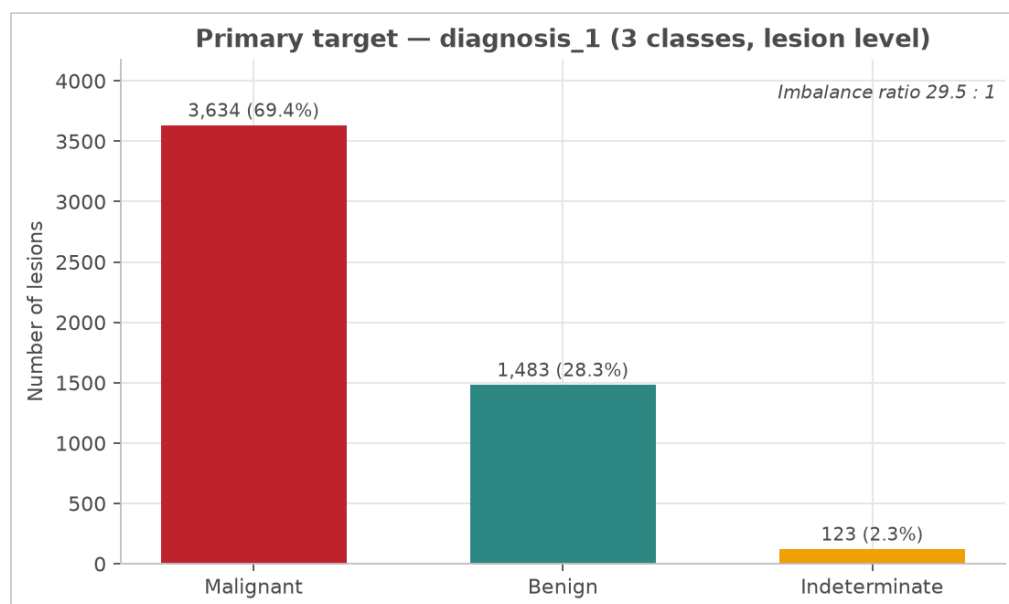
Folder structure on the left, the EDA notebook open.



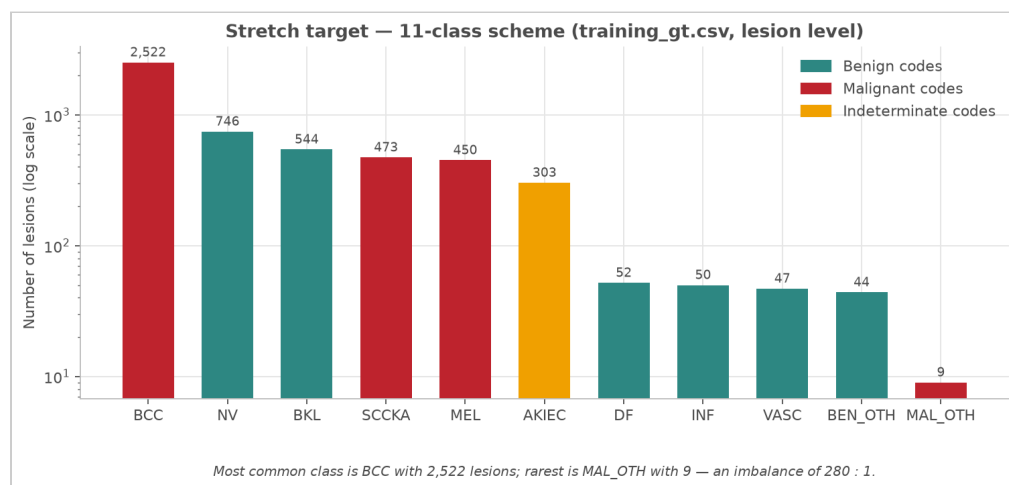
Same project with src/milk10k/data.py open — the loading and structure checks.

What the EDA showed

- 69.4% of lesions are malignant. Always predicting "malignant" gives 69.4% accuracy, so accuracy alone says nothing.
- Every lesion has exactly two images. A random split by row can put both on opposite sides of train and test, so the split has to group by `lesion_id`.
- The 11-class target is very unbalanced: BCC has 2,522 lesions, MAL_OTH has 9.
- AKIEC is split between Indeterminate (123) and Malignant (180), so the 11-class label cannot be translated into the 3-class one with a lookup table.
- 61% of lesions are a single skin tone class.
- Median age is 65 and 60% are men — a specialist clinic population, not the general public.
- Images are not all the same size, so they need resizing before training.

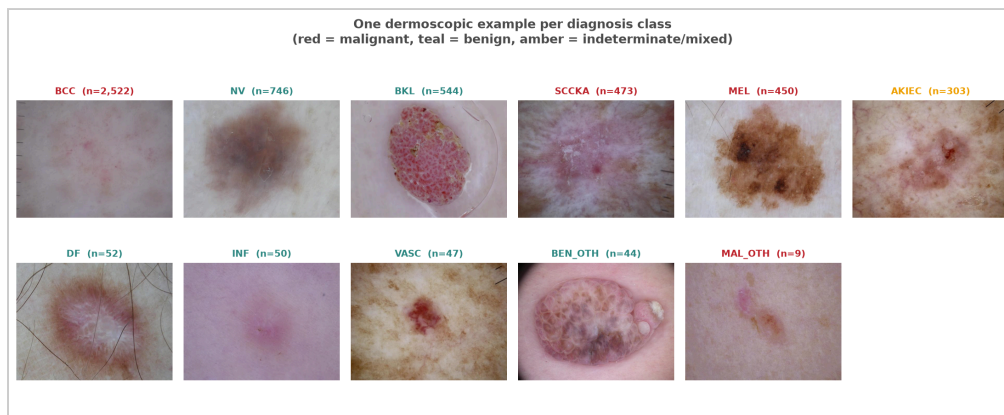


The 3-class target. Malignant dominates — the opposite of the general population.

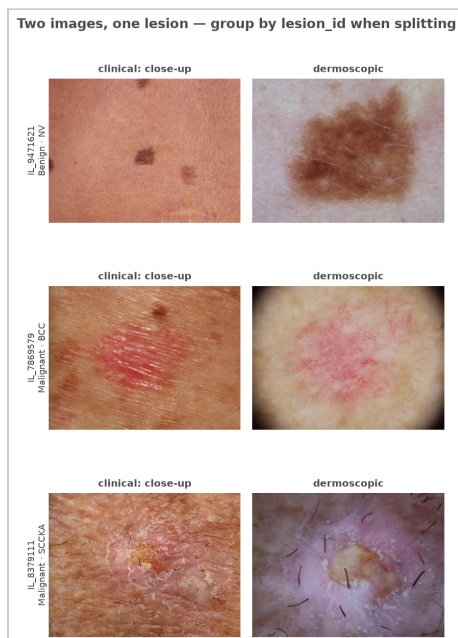


The 11-class target, log scale. 280:1 between the largest and smallest class.

The images

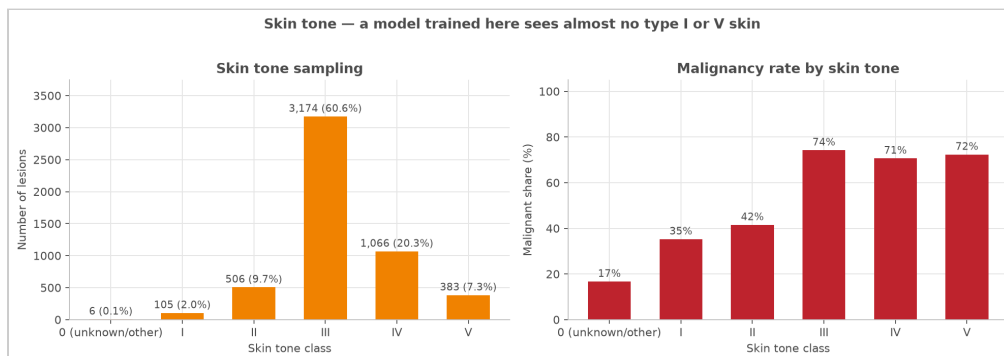


One dermoscopic example per class. BCC, the most common cancer here, is often just a pale pink patch, while a benign mole looks more dramatic.



The same lesion in both modalities. This is why the split has to group by lesion_id.

Who is in the data



Skin tone sampling and malignancy rate. One class is 61% of the data, so overall scores would hide how the model does on the rare tones.